

Very-large-mode-area photonic bandgap Bragg fiber polarizing in a wide spectral range

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A design of a polarizing all-glass Bragg fiber with a microstructure core has been proposed for the first time. This design provides suppression of high-order modes and of one of the polarization states of the fundamental mode. The polarizing fiber was fabricated by a new, simple method based on a combination of the modified chemical vapor deposition (MCVD) process and the rod-in-tube technique. The mode field area has been found to be about $870\ \mu\text{m}^2$ near $\lambda = 1064\ \text{nm}$. The polarization extinction ratio better than 13 dB has been observed over a 33% wavelength range (from 1 to $1.4\ \mu\text{m}$) after propagation in a 1.7 m fiber piece bent to a radius of 70 cm. © 2011 Optical Society of America

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The increase of the average and peak output power of fiber lasers and amplifiers requires utilization of large-mode-area (LMA) fibers, which allow suppression of the nonlinear effects. Another requirement for most high-power fiber lasers and amplifiers is linear polarization of the output beam. To meet this requirement, LMA fibers should be polarization-maintaining (PM). Fibers able to polarize light, the so-called PZ fibers, are more preferable, because they simplify the design of linearly polarized fiber laser devices, ease the alignment of the polarization axes of the fibers being spliced, and ensure a high polarization extinction ratio (PER) over a broad bandwidth. A PZ fiber differs from a PM fiber in that there is an “unguided” polarization state of the fundamental mode that has significantly higher optical loss compare to another (“guided”) one.

The widest single-polarization window, from $\lambda \sim 750\ \text{nm}$ to $1250\ \text{nm}$ (i.e., normalized bandwidth $\Delta\lambda/\lambda > 50\%$) has been obtained in an LMA photonics crystal fiber (PCF), with fundamental mode field area being $700\ \mu\text{m}^2$ [1]. The largest fundamental mode field area ($2300\ \mu\text{m}^2$) has been also demonstrated in a PCF, but its polarization bandwidth was lower, $\Delta\lambda/\lambda \sim 5\%$ ($1030\text{--}1080\ \text{nm}$) [2]. The highest PER ($>60\ \text{dB}$) was achieved in a step-index fiber having two holes, but only in a 3% spectral range and mode field area $\sim 30\ \mu\text{m}^2$ [3].

The main disadvantage of the PCF is the presence of air holes in the fiber cross section, which complicates or even makes impossible (in the case of very LMA PCFs) the splicing with other fibers. All-solid single-polarization fibers are free from this disadvantage; however, they have so far yielded much worse results. The widest polarization bandwidth ($\Delta\lambda/\lambda \sim 13\%$) was achieved in the fiber design based on W-profile (the fast polarization of the fundamental mode was cut off) with fundamental mode field area being $26\ \mu\text{m}^2$ ($\lambda = 840\ \text{nm}$) [4]. The fundamental mode field area of about $50\ \mu\text{m}^2$ (estimated from the core dimension) was achieved in the all-solid

hybrid (photonic bandgap guidance assisted by total internal reflection) fiber with polarization bandwidth of $\Delta\lambda/\lambda \sim 2.4\%$ near $1166\ \text{nm}$ [5].

A promising all-solid LMA fiber type is the photonic bandgap Bragg fiber. Its cross section consists of a core with a low refractive index surrounded by cylindrical layers (called Bragg mirror) with alternating high and low refractive indices. Light is confined in the core owing to coherent Fresnel reflection from the Bragg mirror.

In this Letter, we report on the design and the fabrication of an all-solid LMA Bragg fiber, which ensures efficient suppression of high-order modes (HOMs) and provides record mode field area and polarization bandwidth for all-solid PZ fibers.

Bragg fibers are, strictly speaking, multimode, but HOMs can be filtered out owing to their higher optical loss as compared to the fundamental mode. However, the LP_{11} mode has a relatively low loss and should be suppressed forcibly. To obtain this, we used the Comsol software and designed a microstructured core in which the LP_{11} mode propagation is strongly affected by some inclusions [see Fig. 1(a)]. The physical principle of the microstructured-core Bragg fiber is as follows: The low-index B-doped rods make the fiber birefringent, define the polarization direction of the eigenmodes and, consequently, the positions of the sidelobes of the LP_{11} mode [Fig. 1(a), left]. If we now add two low-index F-doped rods right in the position of the sidelobes, the shape of the LP_{11} mode field will be significantly distorted (compare the contour plots in Fig. 1(a), right and left). The joint effect of the rods and of the Bragg mirror causes the LP_{11} mode to leak away into the cladding, which means that its loss will strongly increase. Thus, one may expect a larger ratio of the light powers propagating in the fundamental mode and HOMs.

It was found that properly chosen design of the Bragg fiber having a microstructured core with B- and F-doped rods [Fig. 1(a) right] could also make the fiber

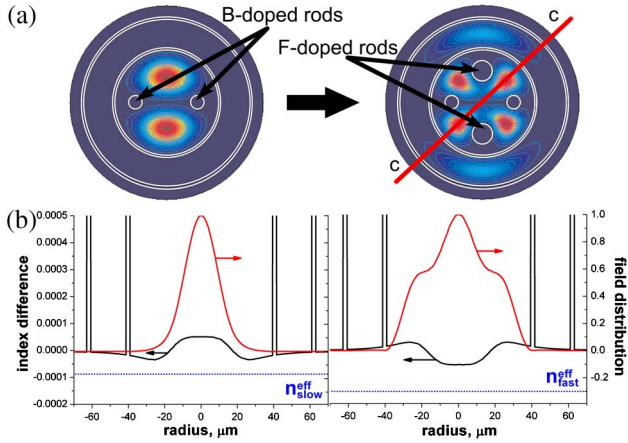


Fig. 1. (Color online) (a) Calculated influence of the presence of B- and F-doped rods in the Bragg fiber core on the LP_{11} mode-field shape shown by contour plots: the effect of two B-doped rods (left) and of two B-doped and two F-doped rods (right). (b) Refractive index profiles (black), effective refractive indices (blue), and mode-field distribution (red) of the slow (left) and fast (right) polarization states of the fundamental mode calculated for the direction of highest leakage (c - c axis in Fig. 1(a) right).

virtually polarizing. Indeed the size of the B-doped rods is small as compared to the core diameter; thus, the stress field is not uniform over the core cross section, and, consequently, the refractive index profile (not only its value) is different for the fast and slow polarization states of the fundamental mode [Fig. 1(b), black]. If the core diameter is large enough (in modeling we have used $D = 80 \mu\text{m}$), the fast and slow polarization states of fundamental mode become sensitive to the variation of the refractive index. It could be seen that the main part of the power of the slow polarization is located in the core center [Fig. 1(b), left], when for the fast polarization, the mode field broadens [Fig. 1(b), right]. Our numerical modeling shows that the fast polarization becomes much more sensitive to the bending as compared to the slow one. Besides, the optical thickness of the low-index layer of the Bragg mirror is different for the fast and the slow polarizations owing to the difference of their effective refractive indices [see Fig. 1(b)]. This fact allows us to adjust the parameters of the Bragg mirror so as to maintain only the slow polarization.

We chose the design with the diameter of the core equal to $80 \mu\text{m}$ and the Bragg mirror composed of two high-index layers with contrast 0.015. For the fiber cross section in Fig. 1(a) right, the calculations predicted birefringence $B \sim 10^{-4}$. In a straight fiber piece at $\lambda = 1.064 \mu\text{m}$, the slow polarization loss was calculated to be 0.02 dB/m, the fast polarization loss being 0.5 dB/m, and the loss of the LP_{11} mode being 5 dB/m. Thus, the suppression of the fast polarization and HOMs were expected to exceed 1 and 2 orders of magnitude, respectively.

The properties of the designed fiber are very sensitive to the refractive indices difference Δn_c between the core and the low-index layer in the Bragg mirror (it defines the optical thickness of the low-index layer). To retain the constructive interference inside the Bragg mirror, Δn_c should be controlled with the accuracy better than $3 \cdot 10^{-5}$, which is unattainable for the standard MCVD

technology. Therefore, a new method of Bragg fiber fabrication based on the combination of the MCVD and rod-in-tube technologies was verified. The preform fabrication process consisted of three stages: fabrication of the microstructured core (stage I) and fabrication of the cladding using a combination of the MCVD and rod-in-tube techniques (stages II–III).

At stage I, holes were drilled in an undoped silica rod (F300 of Heraeus), then B- and F-doped rods were inserted into the holes, and the assembly was fused to yield the final core rod. At stage II, a GeO_2 - SiO_2 layer was deposited on the inner wall of an F300 tube by the MCVD technology. Then this tube was jacketed over the core rod providing a high-index layer of the cladding (GeO_2 - SiO_2 layer) and a low-index layer (Heraeus F300 silica tube). Finally, at stage III, a second MCVD-produced tube with a GeO_2 - SiO_2 layer was jacketed over the preform prepared at stage II.

Particular attention at stages II and III was paid to the concentricity of the tubes in order not to disturb the constructive interference. Note that the core and the low-index cladding layers are produced from the same material (F300 silica of Heraeus) and, hence, their refractive indices coincide exactly.

The thickness of the outer layer of the Bragg mirror was chosen so as to provide a coherent reflection from the silica-polymer boundary [6].

A tomography picture of the final preform is shown in the inset of Fig. 2(a). A fiber with a cladding diameter of $175 \mu\text{m}$ and a core diameter of $80 \mu\text{m}$ was drawn from this preform. The fiber refractive index profile is shown in Fig. 2(a). The sections of the fiber fabricated at stages I–III are shown by the dashed lines.

The near-field distributions at the fiber output, measured by a CCD camera, reveal efficient suppression of HOMs after propagation in a straight fiber piece

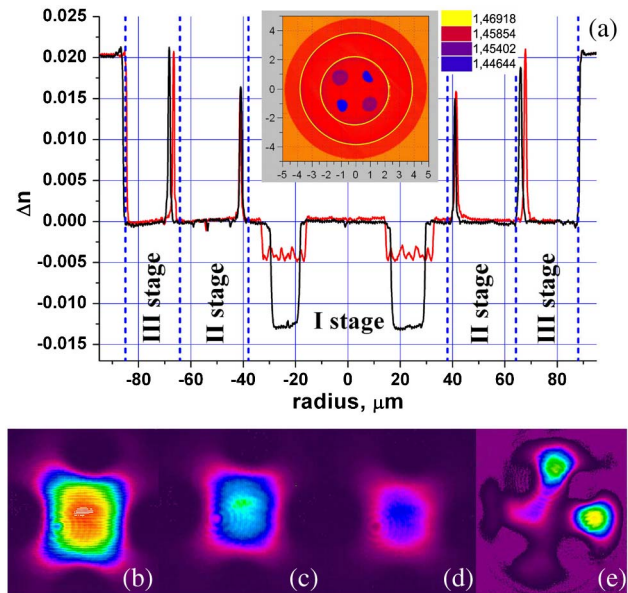


Fig. 2. (Color online) (a) Fiber refractive index profile measured in two orthogonal projections. Inset, preform tomography; (b)–(d) near-field images of the slow polarization of the fundamental mode under different excitation conditions; (e) near-field image of the fast polarization of the fundamental mode.

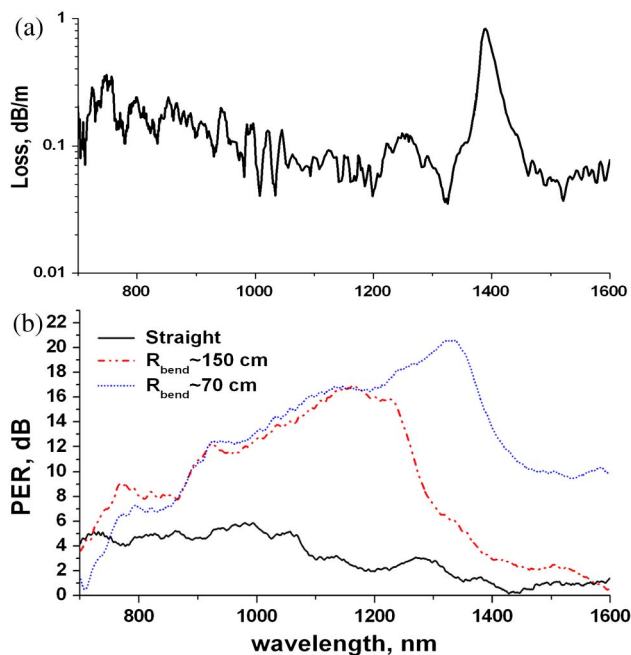


Fig. 3. (Color online) (a) Optical loss of the slow polarization state measured in a straight fiber piece; (b) PER variation with wavelength for a straight fiber and a fiber bent to radii of 70 and 150 cm.

2.7 m in length. Singlemodedness was confirmed by absence of tangible alterations of the near-field distribution of the fundamental mode upon shifting the excitation beam from the Bragg fiber axis. Change of the near-field image of slow polarization upon shifting the excitation beam along the c - c axis is shown in Figs. 2(b)–2(d). The mode field area (at the $1/e$ level) of the slow polarization was estimated to be as large as $870 \mu\text{m}^2$, which corresponds to an average mode-field diameter (MFD) of $33 \mu\text{m}$. No distortion of the mode-field shape of the slow polarization was observed at $\lambda = 1064 \text{ nm}$ at bend radii larger than 25 cm. On the contrary, the bend sensitivity of the fast polarization was found to be extremely high: even slight accidental bending of a straight fiber piece led to a significant distortion of the mode shape [Fig. 2(e)]. The fiber was found to be very sensitive to microbending: some sharp bending or point external stress drastically changed the shape of the slow and the fast polarization state of the fundamental mode and even the entire mode composition. A similar effect was observed earlier in [7].

The optical loss of the slow polarization was measured by the cutback technique in a straight fiber piece, $\sim 6 \text{ m}$ in length, to be less than 0.1 dB/m at $\lambda = 1064 \text{ nm}$ [Fig. 3(a)]. The difference between the calculated (0.02 dB/m) and the experimental ($\sim 0.1 \text{ dB/m}$) values could be caused by some sag of the fiber during the measurement. The splicing loss with an ordinary step-index fiber with an MFD of $\sim 20 \mu\text{m}$ did not exceed 2 dB and was caused, mainly, by the MFD mismatch between the two fibers. This loss could be significantly reduced by use of a fiber micro-optics [8].

Propagation of several HOMs with the excitation maxima between the rods and the first high-index layer of the Bragg mirror was observed in short fiber pieces.

These HOMs together with the fast polarization of the fundamental mode strongly attenuated after propagation in a 1.7 m long fiber piece bent to a radius of 70 cm . In this case, the PER exceeded 13 dB at $\lambda = 1064 \text{ nm}$. The PER spectral dependence for various bend radii is shown in Fig. 3(b). One can see that the polarizing window and the PER noticeably increase with the decreasing bend radius. A PER better than 13 dB was obtained in the spectral range from 1 to $1.4 \mu\text{m}$ ($\Delta\lambda/\lambda \sim 33\%$) at a bend radius of about 70 cm . The short wavelength limit of the polarizing window is caused by the emergence of HOMs. The long wavelength limit is due to a reduction of the bend sensitivity of both polarization states. The loss of the slow (fast) polarization at $\lambda = 1064 \text{ nm}$ in a fiber piece bent to a radius of 70 cm was about 1 dB/m (8 dB/m).

For the first time, a design of an all-solid single-mode polarizing LMA Bragg fiber was proposed and realized. The presence of low-index B- and F-doped rods in the core and a proper choice of the Bragg mirror parameters provided efficient suppression of HOMs and of the fast polarization state of the fundamental mode. In fact, bending of a 1.7 m long fiber piece to a radius of about 70 cm led to a single-polarization propagation with a PER better than 13 dB in a 33% spectral range. The mode field area of the slow (guided) polarization of the fundamental mode was found to be about $870 \mu\text{m}^2$ ($\lambda = 1.064 \mu\text{m}$). Thus, to the best of our knowledge, we have demonstrated the highest mode field area and the widest polarizing spectral range ever reported for all-solid polarizing fibers. We have proposed a new, easy, and reproducible fabrication method of such fibers.

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